Streamflow

Katie Markovich, EPS 522/ ENVS 423L (Fall 2025)

Housekeeping

- Problem sets #1-4 will be returned asap this week
- PS #5 assigned Thursday, but not due until October 16
- Streamflow concepts will not be on Midterm, but will be on Final
- Guest lecture on Thursday followed by exam review
- Midterm exam next Tuesday
- You will need a basic calculator for the exam!

Latent and Sensible Heat

Latent Heat (LE): energy absorbed or released as part of a phase change (condensation releases energy, evaporation absorbs energy). Called latent because it's **hidden**, i.e., cannot be measured, like temperature.

$$E = K_E \cdot v_a \cdot (e_s - e_a)$$

$$LE = \rho_w \cdot \lambda_v \cdot E$$

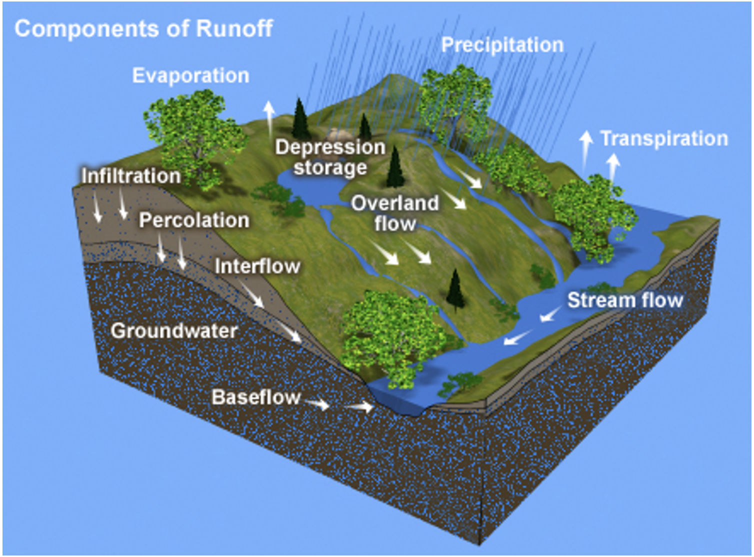
Sensible Heat (H): Energy exchange that you can **sense** because it leads to a change in temperature.

$$H = K_H \cdot v(z_m) \cdot [T_s - T(z_m)]$$

Learning Objectives

1. Stream response to water input
2. Basics of stream gaging
3. Hydrograph separation purpose and methods
4. Runoff-generation mechanisms
5. Stream network and routing
6. Rainfall runoff-modeling

Runoff-Generation Mechanisms



1. Channel precipitation
2. Overland flow
3. Subsurface event flow

Channel Precipitation

Precipitation that falls directly on stream channel.

$$W_{cp} = W \cdot A_{cn}$$

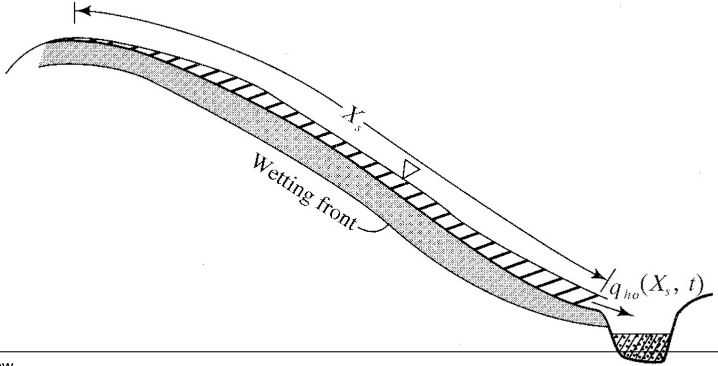
where W is the average water input rate across the drainage area and A_{cn} is the area of the channel network, typically a small fraction (1%) of the total drainage area. Despite the small area, it can be a significant component of q_{pk} and event flow.

Hortonian Overland Flow

"Saturated from above" or "infiltration-excess" overland flow.

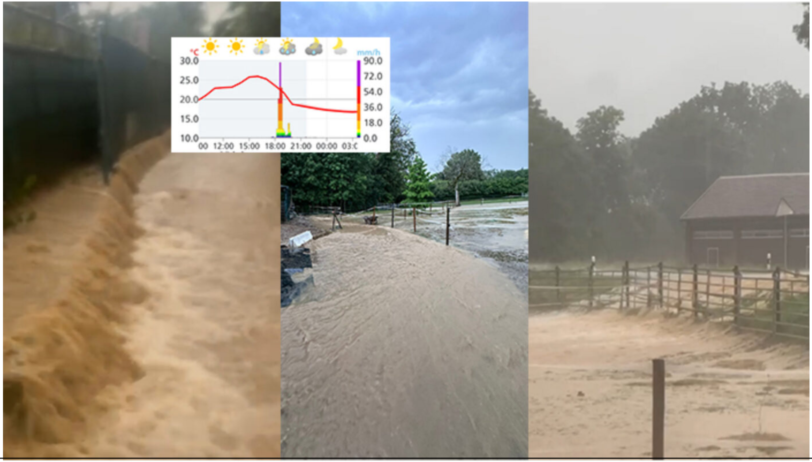
$$q_{ho}(t) = W_{eff}(t) - f(t)$$

where $f(t)$ is the infiltration rate, governed by the saturated hydraulic conductivity (K_h^*) of the surface layer.



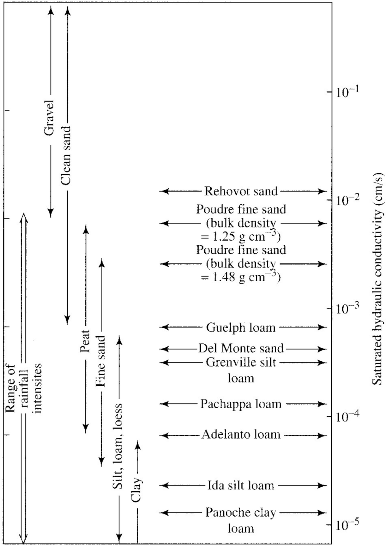
Hortonian Overland Flow

Hortonian overland flow was thought to be the principal runoff-generating mechanism from the 1930s through the 1950s, but it turns out infiltration-excess runoff only occurs in situations of intense rainfall or arid systems.

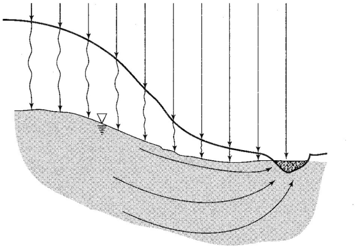


Wait, what is hydraulic conductivity?

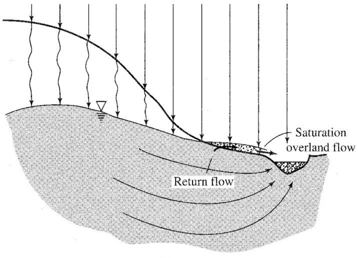
Hydraulic conductivity (K), having units of L/T, is the ability of a porous medium to transmit water. **Saturated** hydraulic conductivity is used for unsaturated porous media, like soils, which can have varying degrees of saturation, and K depends nonlinearly on saturation.



Saturation Overland Flow



(a)



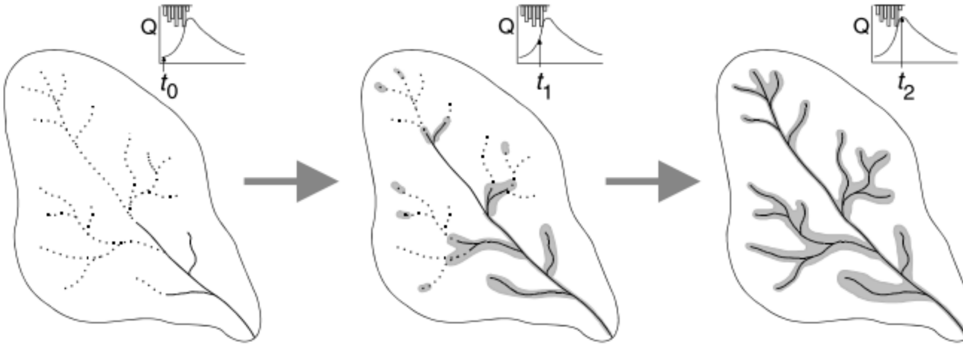
(b)

Also called Dunnian overland flow, where water input infiltrates and saturated zone "breaks out" of the land surface.

First observed in humid regions with shallow water table and gaining streams.

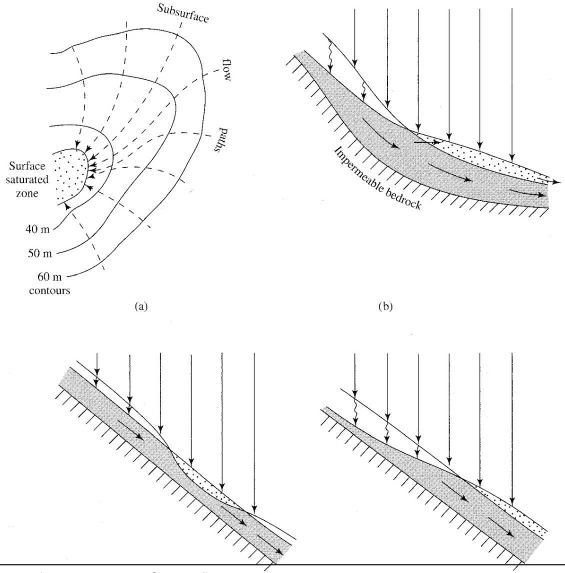
Variable Source Area (VSA)

A concept that saturation excess is the dominant contributor to runoff and extent of areas saturated from below varies widely, dependent on antecedent wetness.



Issues?

Mechanisms for Saturation Excess

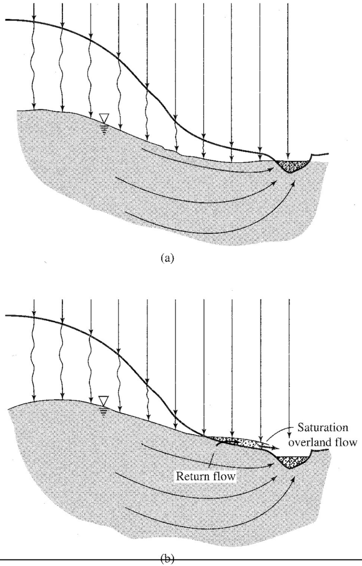


- a. convergence
- b. slope break
- c. local thinning out of soil
- d. saturation excess

Subsurface Event (Storm) Flow

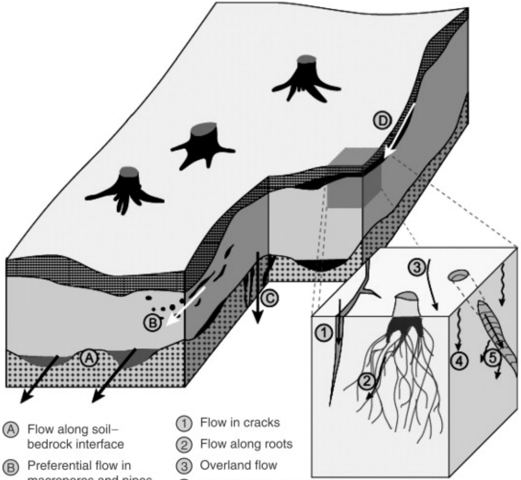
- Subsurface contribution to the response hydrograph that is hydraulically distinct from baseflow.
- Often referred to as **interflow**
- Can occur in the **saturated** and/or **unsaturated** zone.

Groundwater Mounding



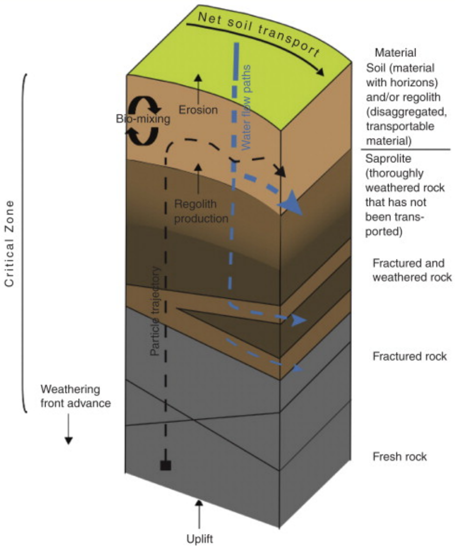
The same process that leads to saturation excess overland flow (infiltration of event water that leads to a rise in the saturated zone) also leads to larger subsurface contributions to streamflow due to steeper hydraulic gradients.

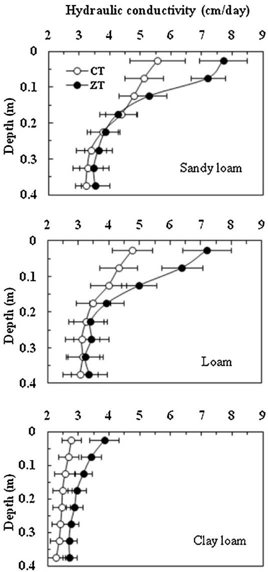
Perched Saturated Zones



Ⓐ Flow along soil–bedrock interface
Ⓑ Preferential flow in macropores and pipes
Ⓒ Flow through bedrock in fissures
Ⓓ Lateral flow in a perched water table between organic and mineral soil horizon
① Flow in cracks
② Flow along roots
③ Overland flow
④ Infiltration into soil matrix
⑤ Flow in earthworm burrows and infiltration into surrounding soil matrix

Hydraulic Conductivity vs Depth





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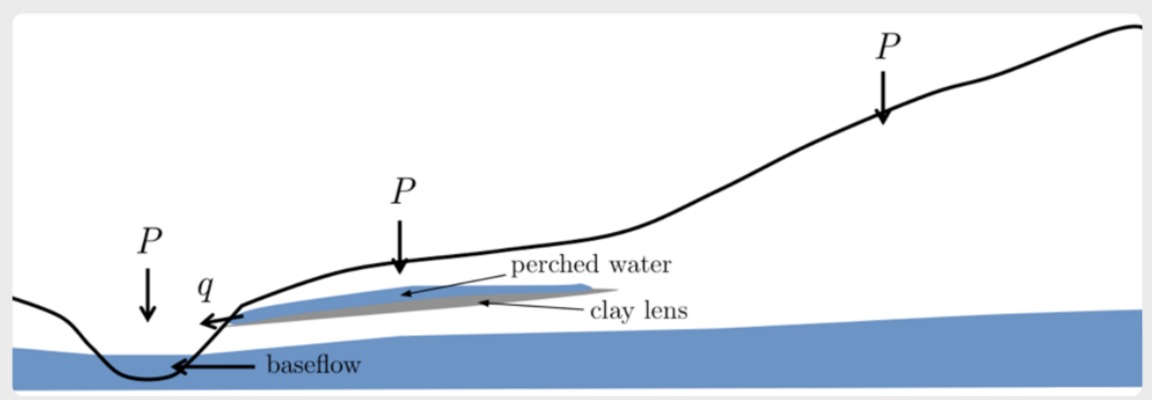
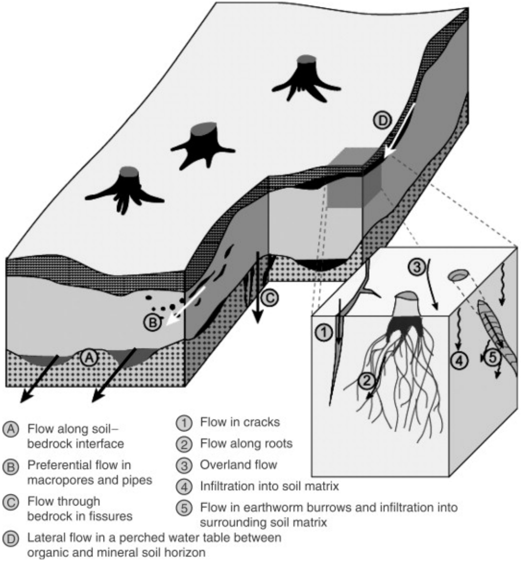
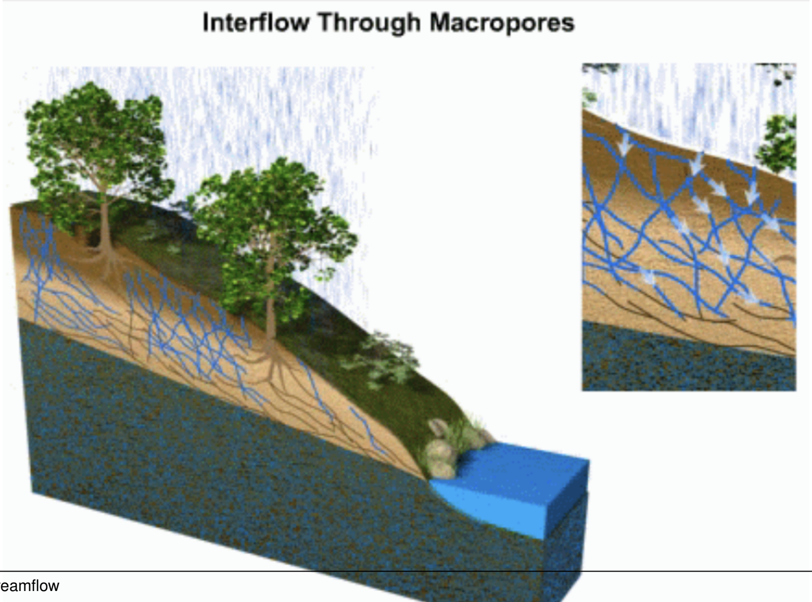


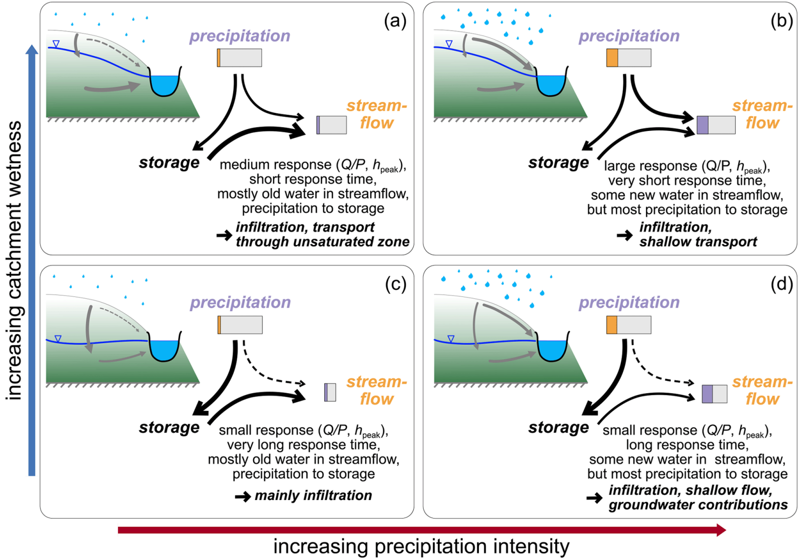
Macropore Flow



Macropore Flow

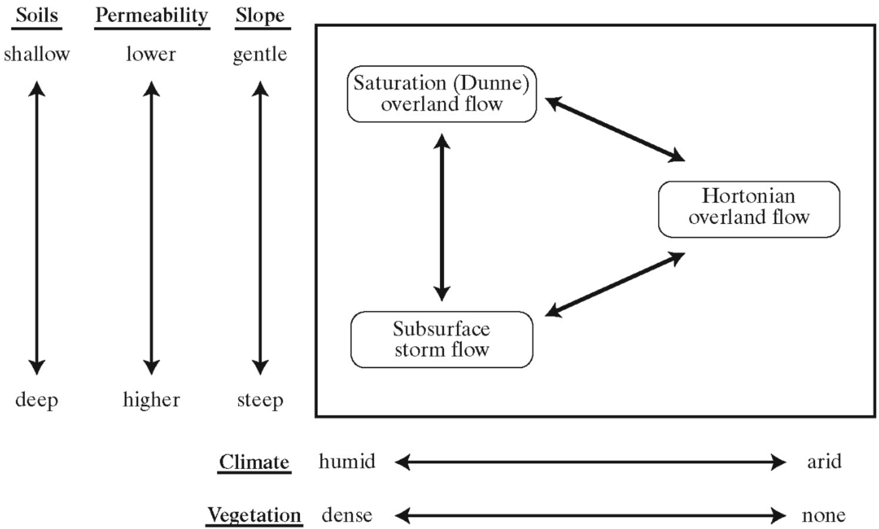


Antecedent Wetness

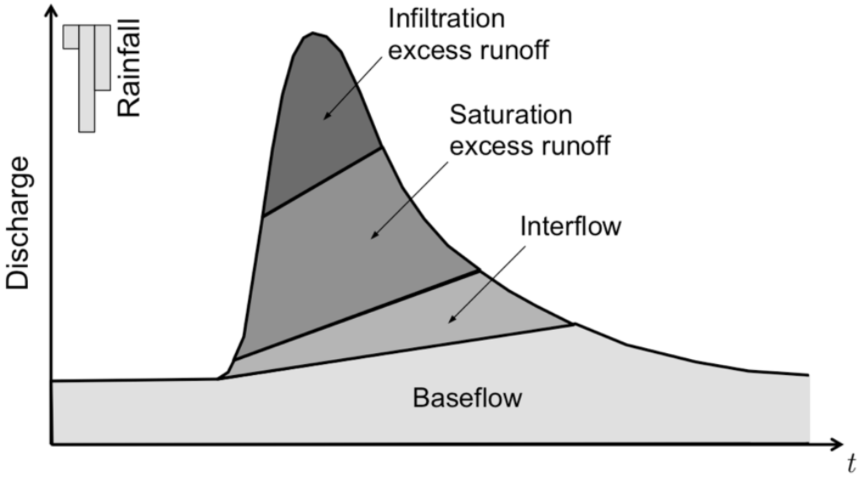


Knapp et al. (2025)

Runoff-generation Mechanisms



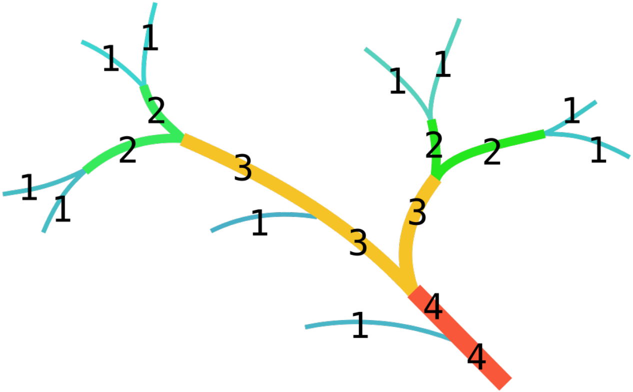
Old Water Paradox, revisited



Stream Network

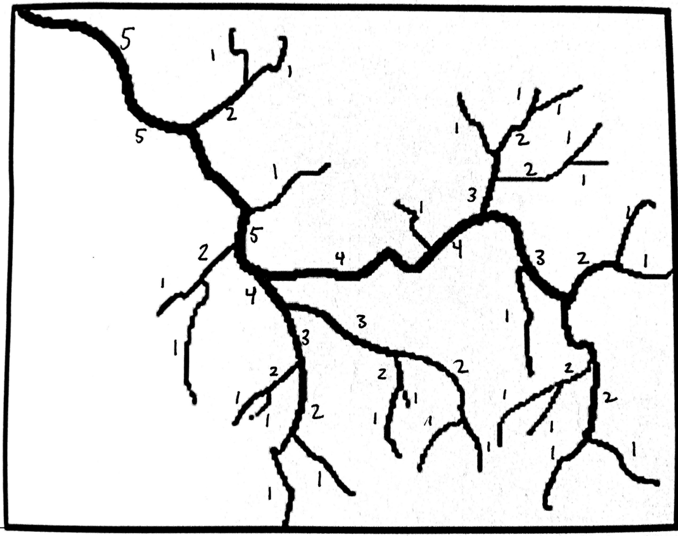
Strahler (1975) stream order is most common framework for quantitatively describing stream networks.

- Tributaries have an order of 1
- Confluence of two 1st order streams is a 2nd order stream, and so on
- Confluence of a lower order stream does not change order



Stream Order Exercise

What is the order of the stream?



Stream Network

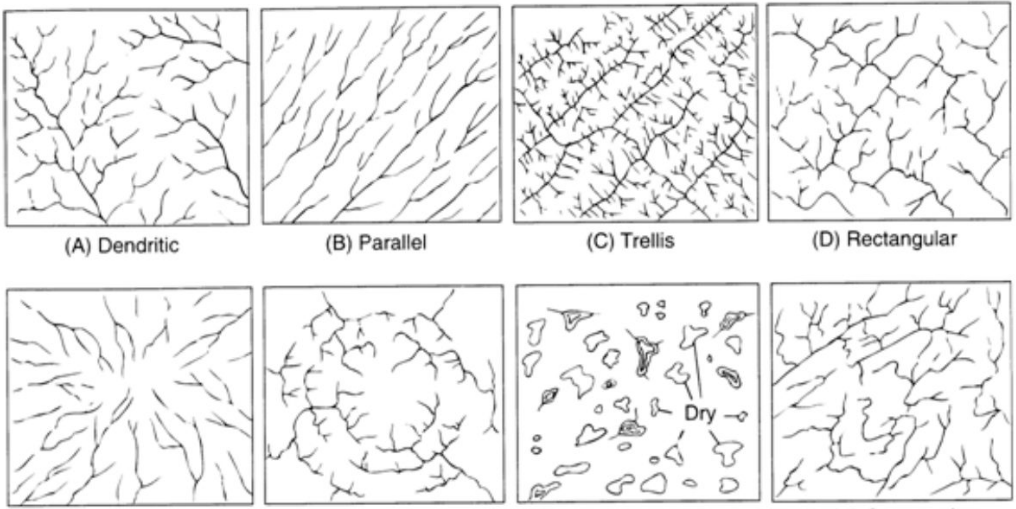
Drainage density is the total length of streams ($\sum L$) draining an area (A_d)

$$D_d = \frac{\sum L}{A_d}$$

Relation of D_d to:

1. average precipitation: higher P, higher D_D
2. soils and bedrock: higher permeability, lower D_D

Stream Patterns



(A) Dendritic (B) Parallel (C) Trellis (D) Rectangular

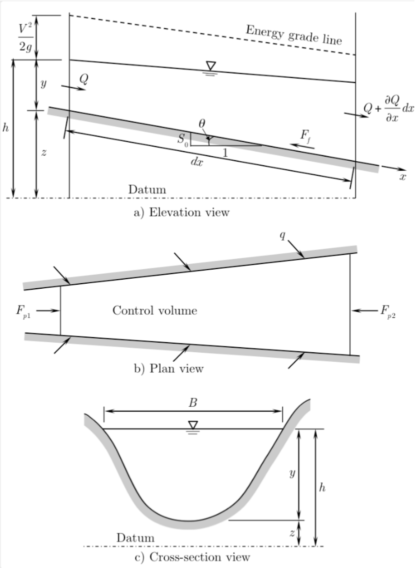
(E) Radial (F) Annular (G) Multibasinal (H) Contorted

Hydraulic Streamflow Routing

We can estimate discharge at any point along a stream channel, assuming flow is effectively 1-dimensional in the x-direction with the continuity equation:

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} - q = 0$$

where A is cross-sectional area, t is time, Q is discharge, and q is inflows/outflows to the stream channel.

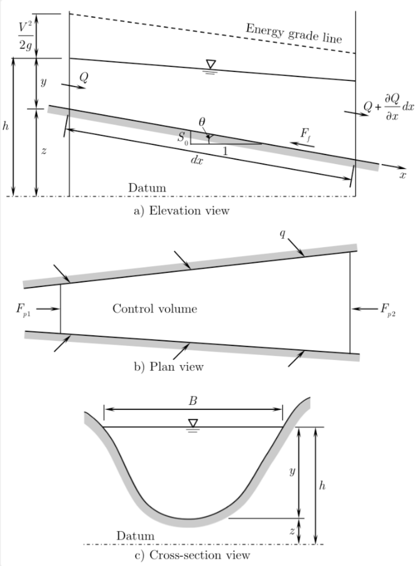


Hydraulic Streamflow Routing

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} - q = 0$$

This is a partial differential equation (PDE), whose solution requires specifying:

- initial conditions (i.e., starting A and Q along the channel)
- boundary conditions (i.e., Q inflows to the channel)



a) Elevation view

b) Plan view

c) Cross-section view

Hydraulic Streamflow Routing

Because the continuity equation has 2 unknowns (Q, A), we have to apply an independent constraint to help solve it. We do so using the conservation of momentum:

$$\frac{1}{A}\frac{\partial Q}{\partial t} + \frac{1}{A}\frac{\partial}{\partial x}(\frac{Q^2}{A}) + g\frac{\partial y}{\partial x} - g(S_0 - S_f) = 0$$

where g is gravitational acceleration, y is water depth (related to A), S_0 is the channel bed slope, S_f is the head loss per unit length of channel

Saint-Venant Equations

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} - q = 0$$

$$\frac{1}{A}\frac{\partial Q}{\partial t} + \frac{1}{A}\frac{\partial}{\partial x}(\frac{Q^2}{A}) + g\frac{\partial y}{\partial x} - g(S_0 - S_f) = 0$$

Together, the continuity and conservation of momentum equations are called the **Saint-Venant Equations** or **Shallow Water Equations** for 1-D, unsteady flow in a channel. These are typically solved numerically by discretizing the channel and approximating ∂ with Δ

Steady flow with no lateral inflow

$$\cancel{\frac{\partial A}{\partial t}} + \frac{\partial Q}{\partial x} - \cancel{q} = 0$$

$$\frac{1}{A}\cancel{\frac{\partial Q}{\partial t}} + \frac{1}{A}\frac{\partial}{\partial x}(\frac{Q^2}{A}) + g\frac{\partial y}{\partial x} - g(S_0 - S_f) = 0$$

Steady, uniform flow with no lateral inflow

$$\cancel{\frac{\partial A}{\partial t}} + \cancel{\frac{\partial Q}{\partial x}} - \cancel{q} = 0$$

$$\cancel{\frac{1}{A}\frac{\partial Q}{\partial t}} + \cancel{\frac{1}{A}\frac{\partial}{\partial x}(\frac{Q^2}{A})} + \cancel{g\frac{\partial y}{\partial x}} - g(S_0 - S_f) = 0$$

$$-\cancel{g} S_0 + \cancel{g} S_f = 0$$

$$S_f = S_0$$

"Head losses from friction balance the elevation head gradient"

Hydraulic Streamflow Routing

For steady, uniform flow with no lateral inflows, the Saint-Venant equations can be simplified to the so-called Mannings Equation:

$$Q = \frac{1}{n} A^{2/3} S_0^{1/2}$$

where n is the Mannings roughness coefficient describing resistance to flow.

Roughness Zone	Shallow Manning's n	Deep Manning's n
Bays and Estuaries	0.060	0.030
Concrete Channel	0.060	0.030
Emergent Aquatic Vegetation	0.300	0.200
Freshwater Marshes	0.300	0.200
Hardwood Conifer Mixed	0.400	0.200
Mangrove Swamps	0.300	0.200
Natural Channel	0.100	0.080
Open Land	0.150	0.040
Paved Road	0.014	0.012
Pine Flatwoods	0.300	0.200
Reservoirs	0.300	0.200
Saltwater Marshes	0.300	0.200
Shrub and Brushland	0.300	0.150
Upland Coniferous Forest	0.400	0.200
Wet Prairies	0.300	0.200
Wetland Forested Mixed	0.400	0.200

Shallow and deep Manning's roughness coefficient (n) for 2D surface model (based on Chow [31] and TR-55 [29]).

Simplifications

$$\frac{1}{A}\frac{\partial Q}{\partial t}+\frac{1}{A}\frac{\partial}{\partial x}\left(\frac{Q^2}{A}\right)+g\frac{\partial y}{\partial x}-g(S_o-S_f)=0$$

Local acceleration term	Convective acceleration term	Pressure force term	Gravity force term	Friction force term

$$\frac{\partial V}{\partial t}+V\frac{\partial V}{\partial x}+g\frac{\partial y}{\partial x}-g(S_o-S_f)=0$$

Kinematic Wave

Diffusion Wave

Dynamic Wave

- Dynamic: good for rapid/abrupt change in conditions like hdyraulic jumps, shock waves, etc. but overkill in most hydro application.
- Diffusive: neglects inertial term and assumes wave spreads diffusively as it travels.
- Kinematic: simplest, neglecting pressure and inertial terms, assumes gravity and friction are dominant forces.

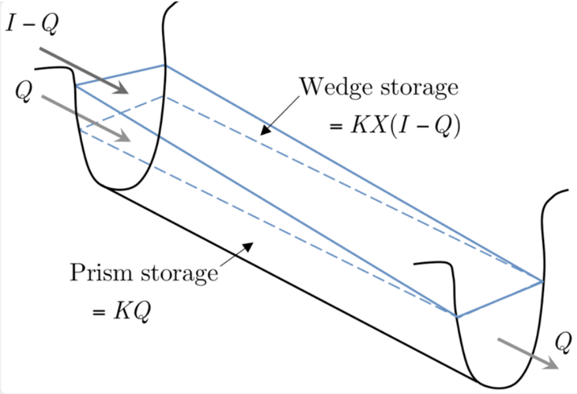
Hydrologic Routing

Conservation of mass at the stream reach scale (i.e., lumped rather than discretized).

$$O_t = \left(\frac{\Delta t - 2KX}{2K(1-X)+\Delta t}\right) I_t + \left(\frac{\Delta t + 2KX}{2K(1-X)+\Delta t}\right) I_{t-1} + \left(\frac{2K(1-X) - \Delta t}{2K(1-X)+\Delta t}\right) O_{t-1}$$

Muskingum Streamflow Routing

where O is outflow, t is time, X is a weighting coefficient (0 to 0.5, typically 0.3 for natural streams) and K is the travel time of a flood wave through the reach.



Rainfall-Runoff Modeling Methods

1. Empirical/Conceptual
2. Physically-based

SCS Curve Number

Soil conservation service (SCS) curve number approach is an empirical approach to estimate precipitation excess (P_e), equivalent to event runoff, for a "normal" wetness condition.

$$P_e = \frac{(P - 0.2 \cdot S_{max})^2}{P + 0.8 \cdot S_{max}}$$

$$S_{max} = \frac{25,400}{CN} - 254$$

where CN is the curve number (0-100) based on hydrologic soil group (A, B, C, D*tied to infiltration capacity) and land cover type.*

Curve Numbers

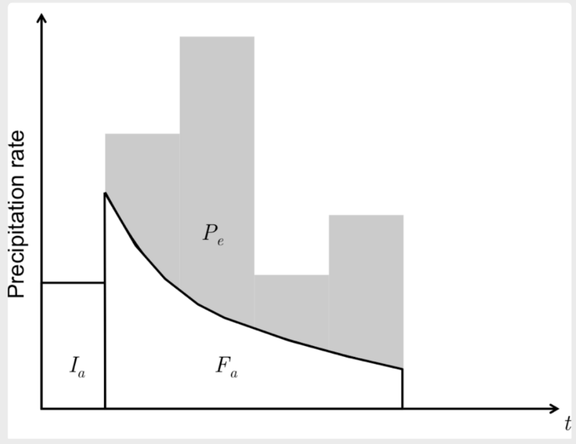
https://www.hec.usace.army.mil/confluence/hmsdocs/hmstrm/cn-tables

What's the curve number?

Assume soil is in good condition and it is a hydrologic soil group B.

Curve Numbers

What kind of runoff-generation mechanism does this resemble? Hortonian Overland Flow

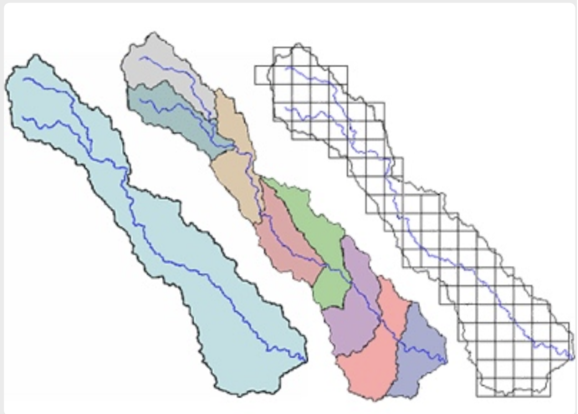


What major factor does this not account for? Antecedent wetness

Physically-based Rainfall-Runoff Modeling

Simulate runoff based on physical processes (such as infiltration, evaporation, unsaturated zone moisture redistribution, recharge, and groundwater flow).

There are a dozen of "physically-based" models available differing in their discretization strategy, process inclusion and representation, etc.



Physically-based Rainfall-Runoff Modeling

Name	Type	Channel Routing	Overland Flow
SWAT	Semi-Distributed	Muskingum or Variable Storage	SCS Curve Number
PRMS	Lumped/Distributed	Muskingum or Mass Balance	Mass Balance (Hortonian, Dunnian, Interflow)
HEC-HMS	Semi-distributed	Muskingum, Kinematic Wave, and many more...	Unit Hydrograph, Kinematic/Diffusive Wave
MODFLOW6	Distributed	Mass Balance, Kinematic/Diffusive Wave	n/a, Kinematic/Diffusive Wave
ParFlow	Distributed	Kinematic/Diffusive Wave	Kinematic/Diffusive Wave

Precipitation-Runoff Modeling System